

Alejandro Zárate Macías

AI Engineer

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Profile

AI Engineer and software developer with nearly 5 years of experience building production AI systems, data platforms, ML APIs, and LLM-powered applications across public sector and industrial environments. Experienced in taking AI solutions from proof of concept to MVP and production, including backend services, data pipelines, databases, deployment, monitoring, and user-facing integrations. Specialized in RAG systems, vector search, agentic AI workflows, MCP servers, and cloud-based AI architectures. Strong hands-on background with Python, FastAPI, PostgreSQL, Apache Airflow, Docker, Linux infrastructure, GCP, AWS, and applied ML/NLP systems. Currently pursuing an M.Sc. in Data Science Engineering focused on optimized Spanish embeddings for Agentic-LLM tool selection and context retrieval.

GenAI / Agentic AI Projects

- **Institutional RAG Assistant:** LLM assistant over IIEG documents and databases using retrieval pipelines, vector search, FastAPI, PostgreSQL and institutional APIs.
 - **Agentic AI Tooling Platform:** Deep Agents-based workflows with planning, skills, document reading, sandbox execution, subagents and MCP integrations.
 - **MCP API Layer:** FastMCP servers exposing internal APIs as AI-consumable tools for assistants and agentic systems.
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Work Experience

AI Developer - Institute of Statistical and Geographic Information of Jalisco (IIEG)

Jan 2026 – Present

- Designed and developed GenAI applications for institutional data access, enabling users to retrieve, analyze, and interact with structured and unstructured government information through LLM-powered interfaces.
- Built RAG-based systems over institutional documents and databases, including retrieval pipelines, vector search workflows, context ranking, and API integrations for enterprise data access.
- Developed agentic AI prototypes using planning-oriented agents, subagents, tool calling, document reading, sandboxed execution, skills, and MCP-based integrations.
- Built MCP servers with FastMCP to expose internal system APIs and data services as AI-consumable tools for assistants and agentic workflows.
- Designed and deployed Python/FastAPI services for AI workflows, ML inference, data access, and integration with internal platforms.
- Integrated AI systems with institutional ETL pipelines, PostgreSQL/PostGIS databases, Linux servers, and cloud infrastructure.
- Developed AI-assisted software delivery workflows using Claude Code, Codex, OpenCode, and GitHub Copilot, including project-level instructions, reusable skills, development commands, review flows, and task-specific coding agents.

Software Project Leader - Institute of Statistical and Geographic Information (IIEG)

Jun 2025 - Jan 2026

- Led development of internal software solutions for government data systems
- Designed and implemented LLM-based information retrieval systems
- Coordinated development and maintenance of ETL pipelines for institutional datasets

- Managed deployment and monitoring of services in Linux and cloud environments

Systems Specialist - Institute of Statistical and Geographic Information (IEG)

Dec 2024 - Jun 2025

- Developed web-based data collection systems used by government institutions
- Collaborated in the development of RAG-based AI agents for internal government tools
- Implemented object detection and OCR models for road infrastructure analysis
- Built and maintained ETL pipelines using Apache Airflow
- Implemented data visualization platforms using Apache Superset

Machine Learning Engineer - Intelica (Short-term assignment)

Mar 2024 - Jul 2024

- Developed model inference APIs using FastAPI and ONNX Runtime
- Implemented MLOps workflows using MLflow and DVC
- Automated training and deployment pipelines using GitHub Actions
- Designed and deployed data labeling pipelines with Label Studio

Machine Learning Engineer - Jalisco State Government

Sep 2023 - Dec 2024

- Developed AI-based data analysis pipelines
- Implemented REST APIs for model inference in production environments
- Designed pipelines for training, evaluation, and experimentation of ML models
- Implemented CNN models for classification and segmentation
- Installed and maintained NVIDIA DGX A100 infrastructure for deep learning workloads

Embedded Software Developer Jr - Bosch

Aug 2021 - Aug 2023

- Provided software support for automotive production projects
- Implemented solutions for production code issues
- Managed software release processes and client deliveries
- Developed automation scripts to improve engineering workflows

Education

- **University of Guadalajara**

Bachelor's degree in Computer Science Engineering

Aug 2018 - Dec 2022

Relevant coursework: Artificial Intelligence, Data Structures, Algorithms, Databases, Software Engineering
Graduated with academic excellence.

- **University of Guadalajara**

M.Sc. in Data Science Engineering — In Progress

Expected: Dec 2027

Thesis Topic: Improving Agentic-LLM systems through Spanish-optimized embeddings for tool filtering and context retrieval.

Skills

- **Programming & Backend:** Python, FastAPI, SQLAlchemy, REST APIs, C/C++, SQL, async services
- **Generative AI & Agentic Systems:** LLMs, RAG, vector databases, semantic search, tool calling, MCP servers, FastMCP, Deep Agents, LangChain, agent workflows, subagents, skills, prompt engineering

- **Machine Learning & NLP:** PyTorch, embeddings, NLP, OCR, computer vision, CNNs, model evaluation, ONNX Runtime
 - **Data Engineering:** Apache Airflow, ETL/ELT pipelines, PostgreSQL, PostGIS, DuckDB, data modeling, structured and unstructured data pipelines
 - **Cloud & Infrastructure:** Google Cloud Platform, AWS, Docker, Linux servers, Terraform, CI/CD, GitHub Actions
 - **MLOps & Observability:** MLflow, DVC, experiment tracking, model deployment, inference APIs, evaluation pipelines
 - **AI-assisted Development:** Claude Code, Codex, OpenCode, GitHub Copilot, project agents, coding skills, reusable commands, AI-assisted code review
 - **Visualization & Analysis:** Pandas, NumPy, Jupyter, Apache Superset, Matplotlib
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Languages

- Spanish – Native
- English – B2 / Professional technical communication